nouPix

GETTING STARTED · V1.1.0

FROM ZERO TO LIGHT, STEP BY STEP.

Install nouPix, draw your fixtures, patch them, play content in zones and layers, store scenes, and run the show from a controller or a phone.

nouPix 1.1.0 · macOS · Windows · Linux · Raspberry Pi
Free and open source · GNU AGPL-3.0 · NOULED.STUDIO
noupix.nouled.com



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START HERE Steps 03 to 17 build one small project from an empty canvas: four bars, a matrix, a ring and a wave, in two zones. Every picture is the real app.

01

INSTALL AND OPEN

GET NOUPIX RUNNING

MACOS

macOS 12 or newer · Apple silicon or Intel

1. Download the **.dmg** for your Mac from the download page.
2. Open it and drag **nouPix** onto Applications.
3. First start: click **Done** on the warning, then **System Settings → Privacy & Security → Open Anyway**.

WINDOWS

Windows 10 or newer · 64-bit

1. Download the **installer .exe**.
2. If SmartScreen stops it: **More info → Run anyway**.
3. Open nouPix from the Start menu. Allow it through the firewall on **private networks**, or node discovery won't find your controllers.

LINUX · PI

Any computer · Python 3.8+ and Chrome or Chromium

1. Get the source and run it:

```
$ git clone https://github.com/nouled-studio/nouPix.git
$ cd nouPix
$ python3 noupix.py
```

2. nouPix opens in its own window. Keep the terminal open: closing it stops the output.

FIRST START nouPix opens with a demo layout, already playing. Look around, then start your own project (step 03).

02

THE WINDOW

EVERYTHING ON ONE SCREEN

1. **Top bar**: project, undo, black or white, remote, MIDI, highlight, master, blackout
2. **Fixtures**: everything on the stage, in groups
3. **Sources**: effects, your shaders, images, video, capture
4. **Tools**: select, line, matrix, circle, path, pen, zones
5. **Stage**: your layout, playing live
6. **Dock**: Scenes · Patch · Nodes · DMX monitor
7. **Right panel**: Inspector · Content · Settings
8. **Status bar**: frame rate, pixels, universes, Art-Net



TIP

Right-click anything for its options. H hides the right panel, F fits the stage.

03

A NEW PROJECT

START FROM AN EMPTY CANVAS

1

Project ▾ → New empty project. It starts with one zone playing Rainbow, so every pixel you draw lights up.

2

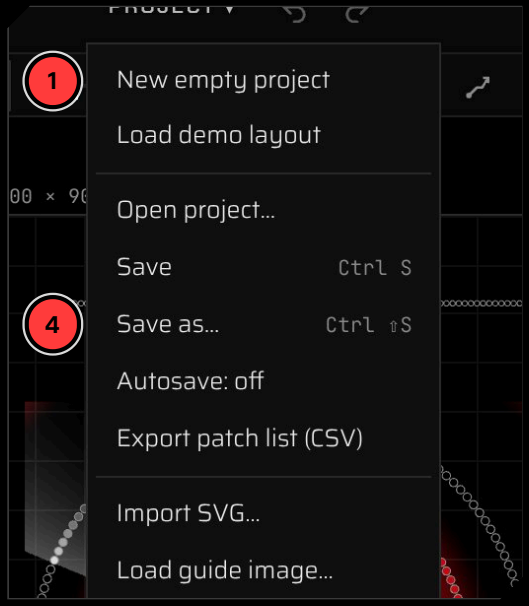
Name it: click the name in the top bar.

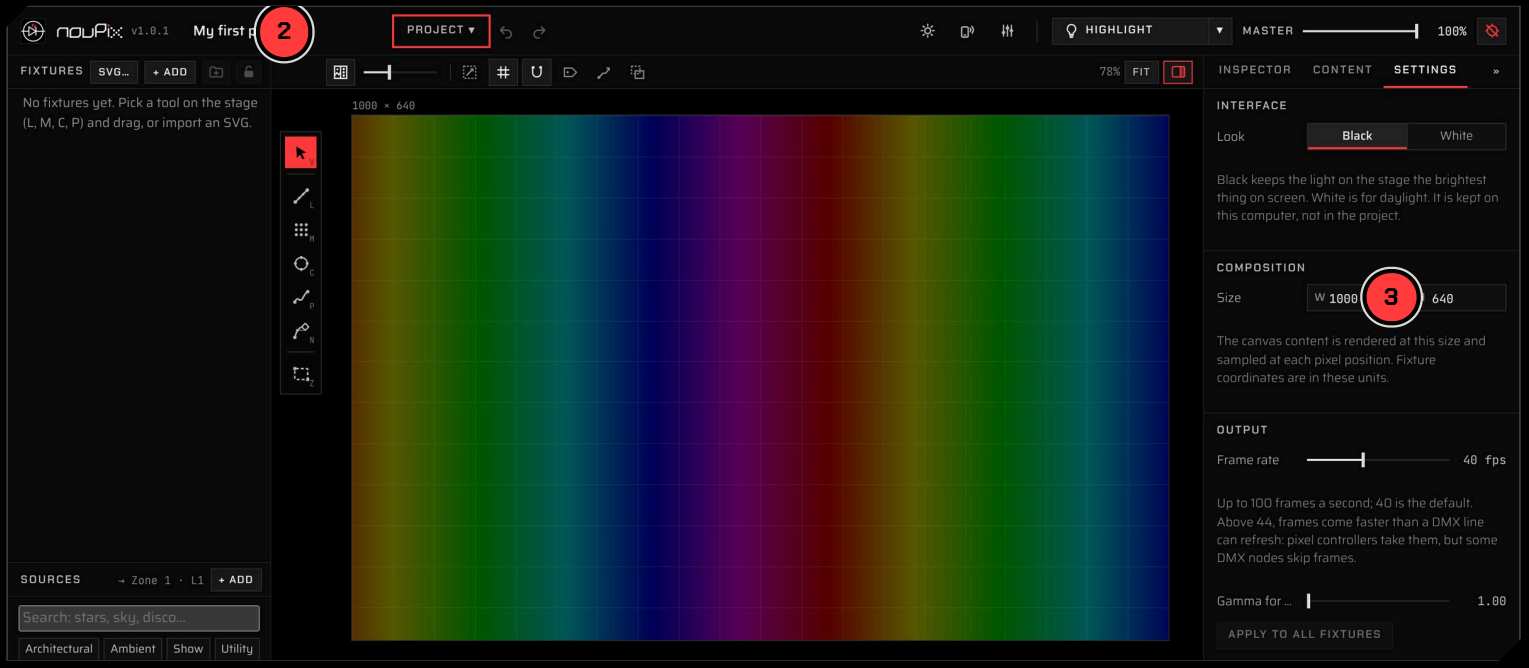
3

Settings → Composition: the size of your canvas, in any unit you like (for example 1000 × 640), and the frame rate.

4

Project ▾ → Save as... writes a **.noupix** file. Turn on **Autosave** and every change is saved to it.





TIP `Ctrl` `S` saves, `Ctrl` `Z` undoes. Projects open with **Project ▾ → Open project...**; the examples on GitHub are ready to open.

04

DRAW FIXTURES

DRAW WHAT YOU BUILT

1

Pick a tool: `L` line or strip, `M` matrix, `C` circle or ring, `P` path, `N` pen for curves.

2

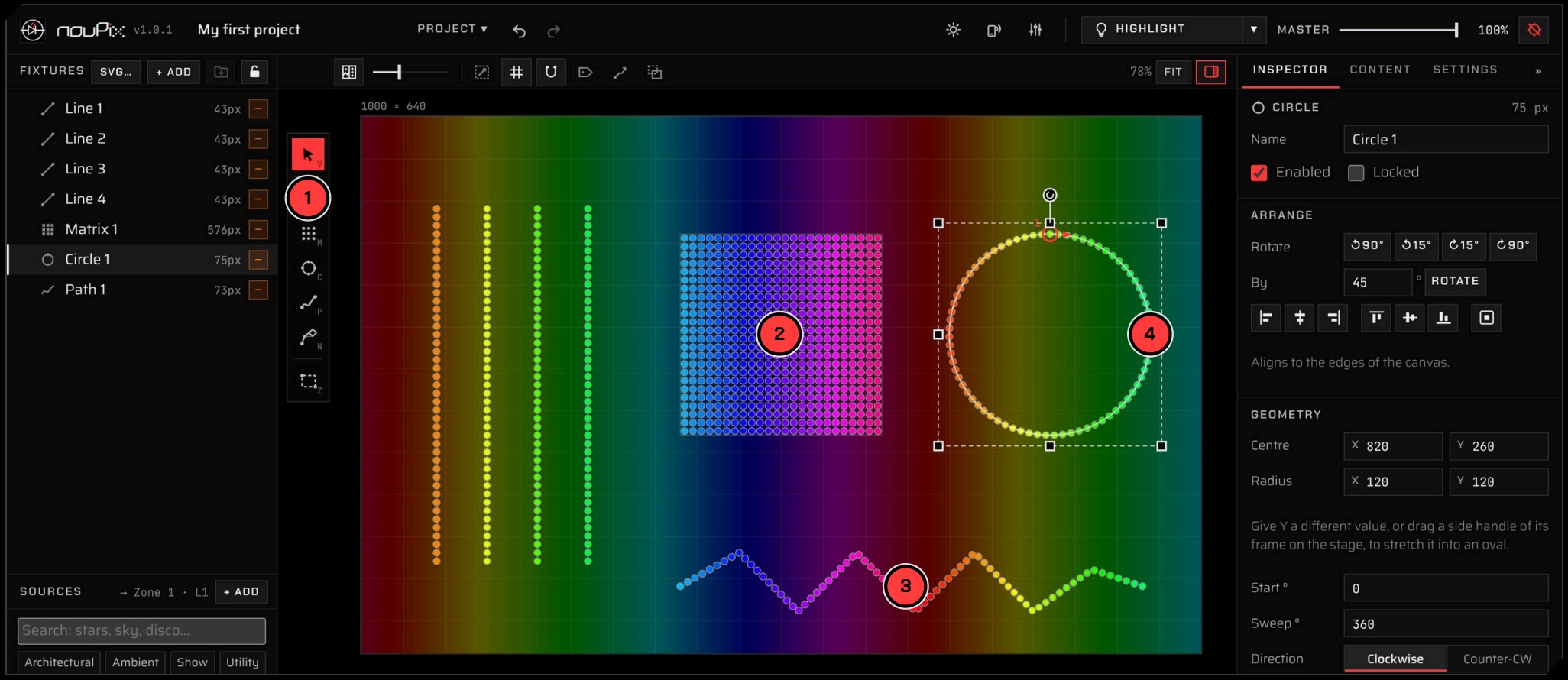
Drag on the stage: a line from end to end, a matrix as a rectangle, a circle from its centre outwards.

3

Path: click the points, then `Enter`. **Pen:** click for corners, drag for curves.

4

`V` selects and moves. `Shift` snaps to 45°, `Alt` turns snapping off. A click with a tool drops a default fixture.



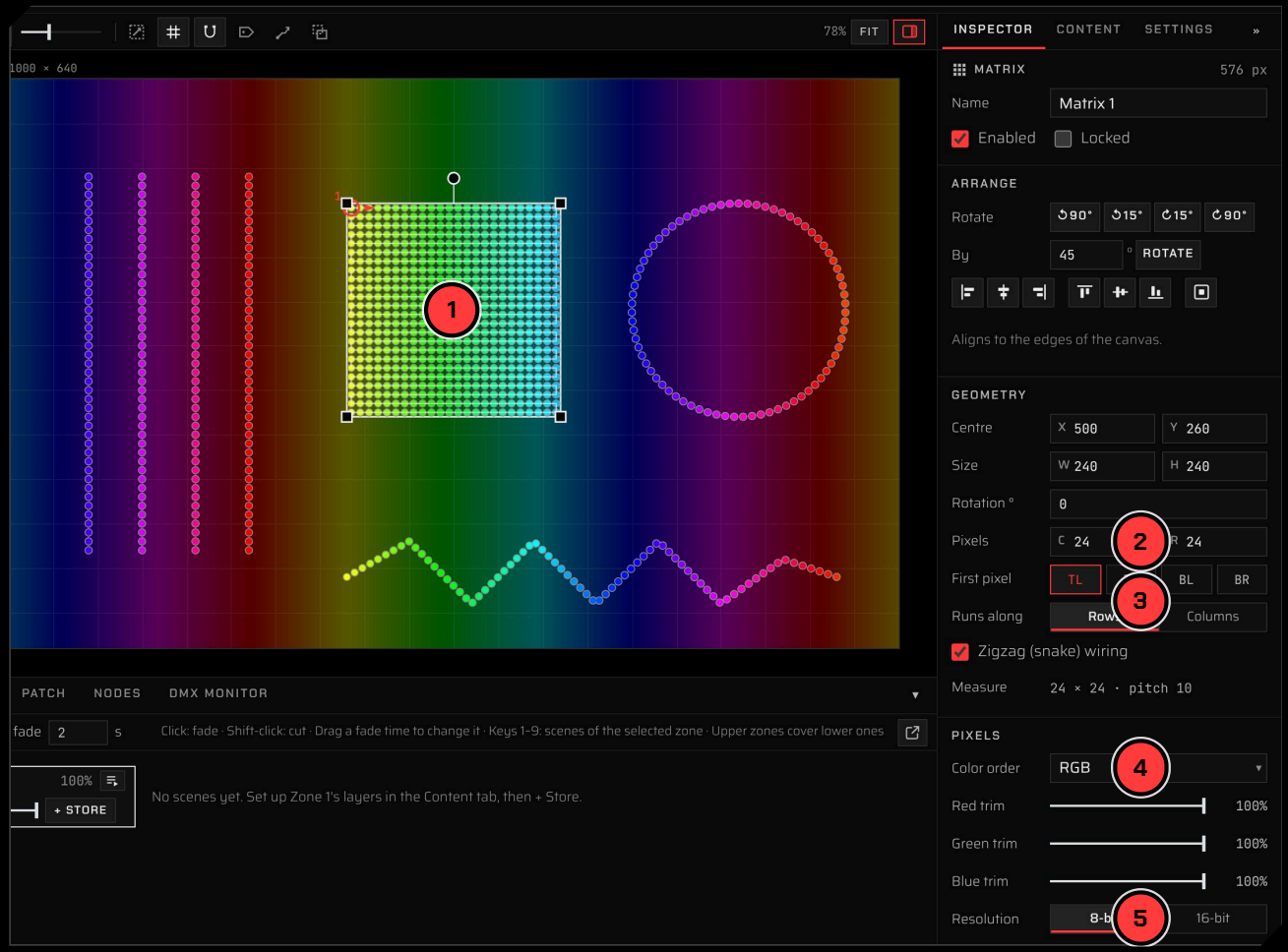
TIP Each new fixture appears in the Fixtures list on the left, with its pixel count.

05

SET THEM UP

PIXELS, WIRING, COLOUR

1. **Select a fixture**: its settings open in the Inspector.
2. **Pixels**: the count, or columns × rows for a matrix.
3. **First pixel and wiring**: which corner it starts from, rows or columns, zigzag.
4. **Colour order** (RGB, GRB, RGBW…) and trims per colour.
5. **Resolution** 8 or 16-bit, and the dimming curve with its graph.



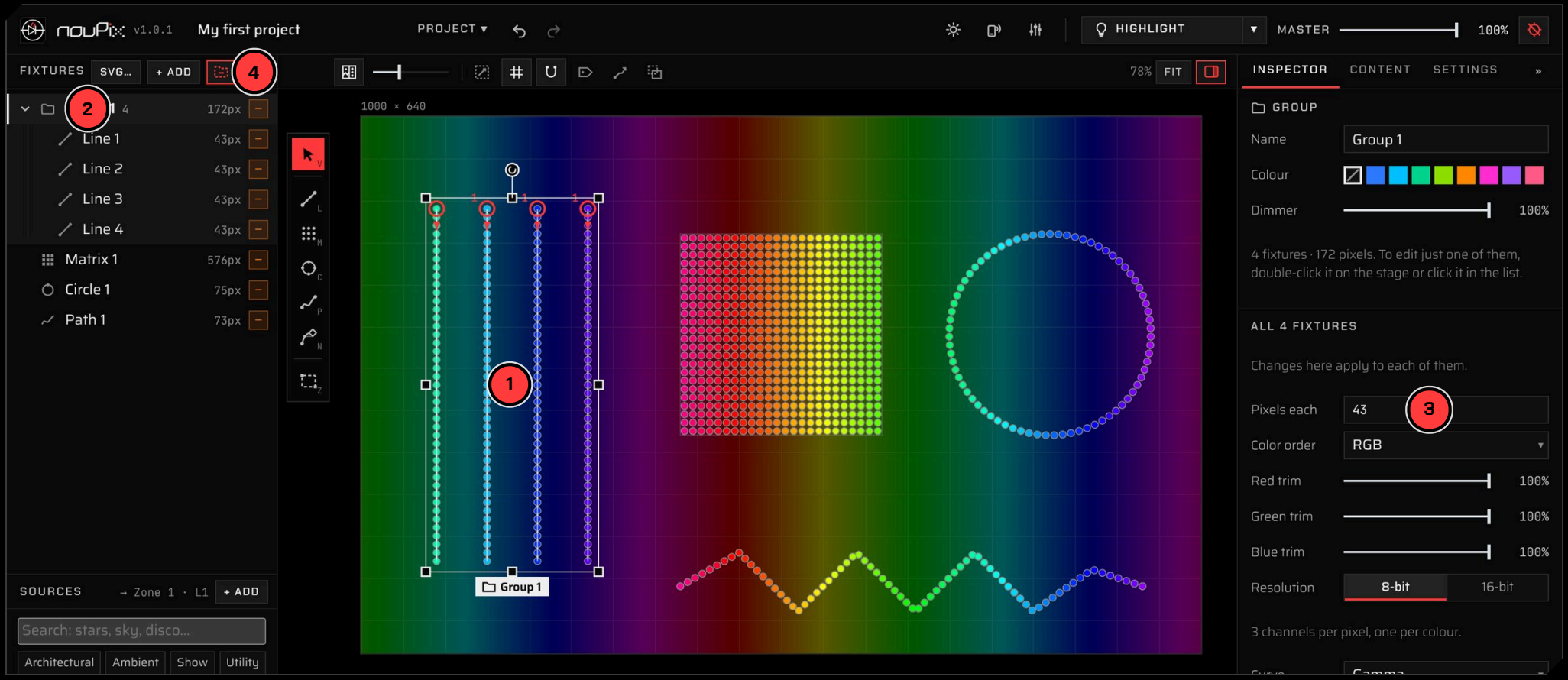
TIP Select several fixtures to set their pixels, colour order or gamma all at once.

06

ARRANGE AND GROUP

KEEP THE LAYOUT TIDY

1. Select several: `Shift`-click, or drag a box around them.
2. Group them with `Ctrl` `G`: the list folds them under one name, with its own dimmer.
3. Rotate, align and set pixels for all of them at once.
4. Lock with `Ctrl` `L` so nothing moves by accident.
5. Copy, paste and duplicate: `Ctrl` `C` `Ctrl` `V` `Ctrl` `D`.



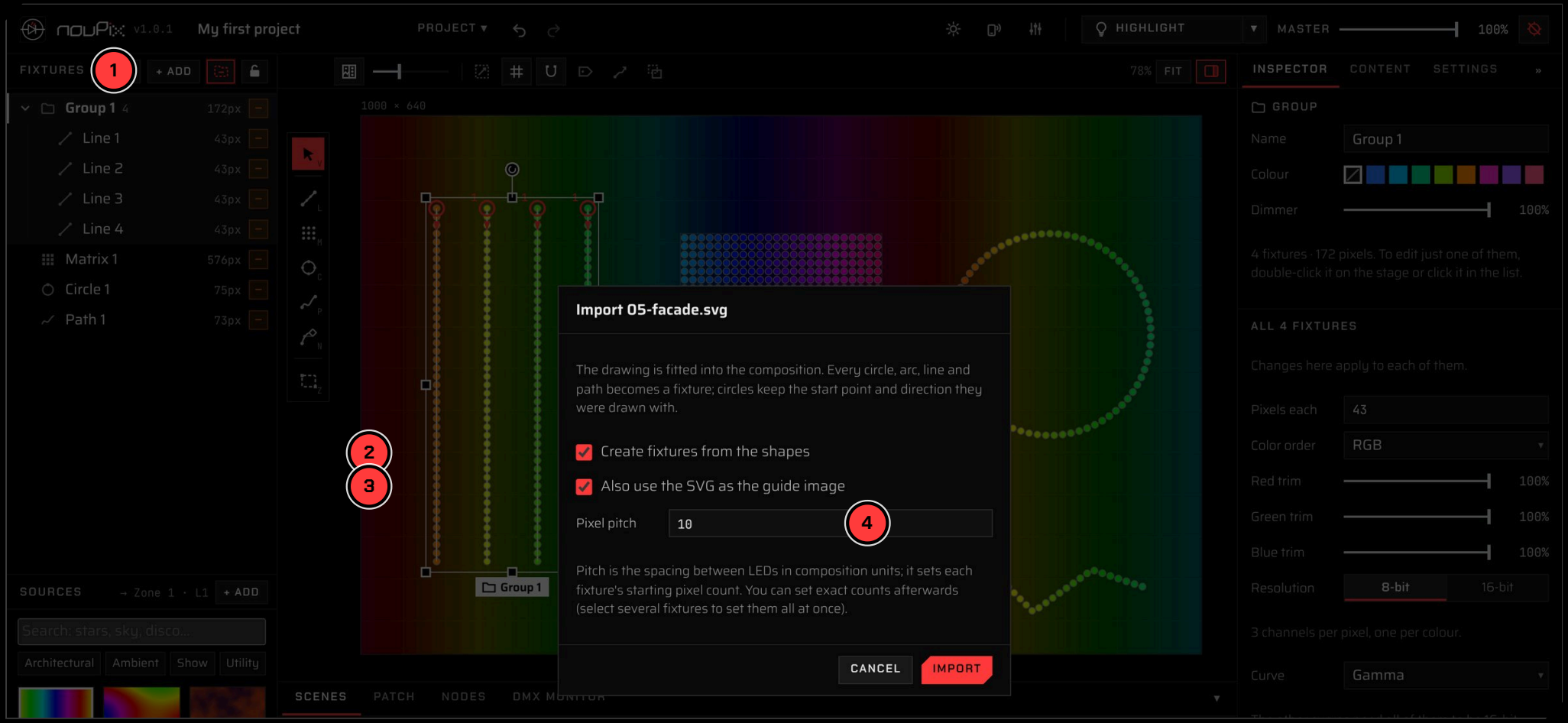
TIP Double-click a fixture inside a group to pick just that one. `R` turns the selection 90°.

07

IMPORT A DRAWING

START FROM YOUR SVG

1. **SVG…** at the top of the Fixtures list, or **Project ▾ → Import SVG…**
2. **Create fixtures**: every circle, arc, line and path of the drawing becomes one.
3. **Use it as the guide image**, under the fixtures, to check them against the building.
4. **Pixel pitch**: the spacing of the LEDs; it sets each fixture's pixel count.



TIP Export the façade from your drawing or 3D software as SVG, one stroke per LED strip.

08

FIND CONTROLLERS

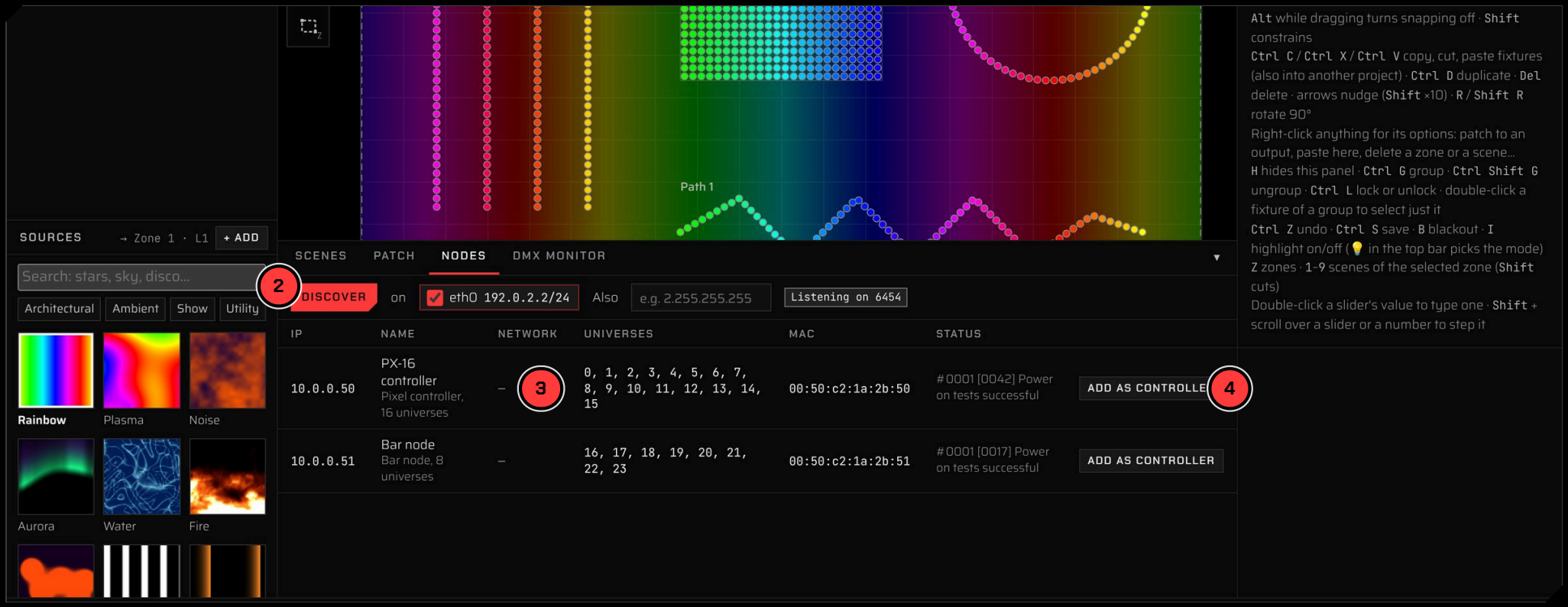
DISCOVER THE NODES

1. Connect the controllers and this computer to the same network, with addresses in the **same range** (for example 10.0.0.x, or 2.x.x.x).

2. **Nodes** tab in the dock → **Discover**: ArtPoll goes out on the networks ticked next to it.

3. Every node that answers is listed, with its **IP**, name and **universes**.

4. **Add as controller**: it goes to the Patch tab with its IP, name and universes, ready to patch.



TIP Not found? Allow nouPix through the firewall on private networks, and give this computer an address in the node's range. A node that doesn't answer ArtPoll can still be added by hand (next step).

09

PATCH

FIXTURES ONTO OUTPUTS

1

The controller is in the **Patch** tab, with the universes it reported (**ArtPoll**).

2

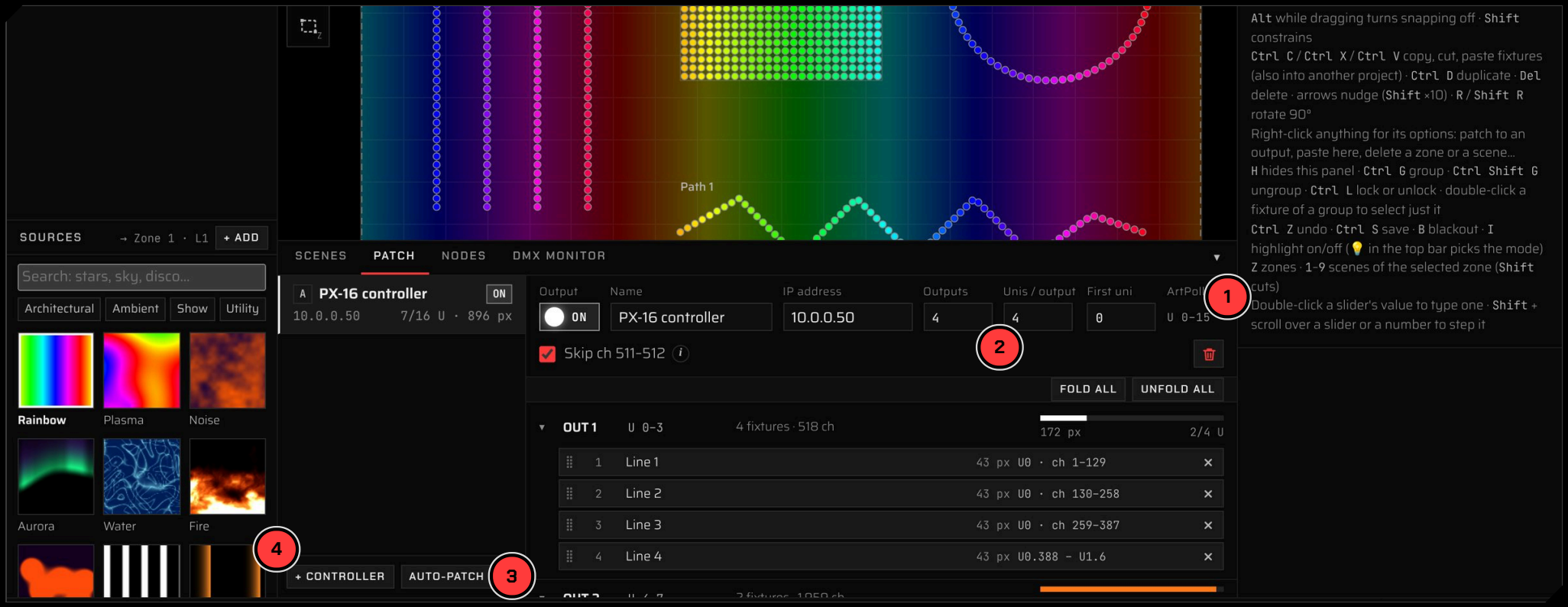
Set **Outputs** and **Unis / output** to how the device is set up: ArtPoll tells the universes, not how they are split over the outputs.

3

Drag fixtures from the list onto an output, or press **Auto-patch**. The order on an output sets the addresses.

4

Not found by ArtPoll? + Controller adds one by hand: type its IP address, outputs and universes per output.



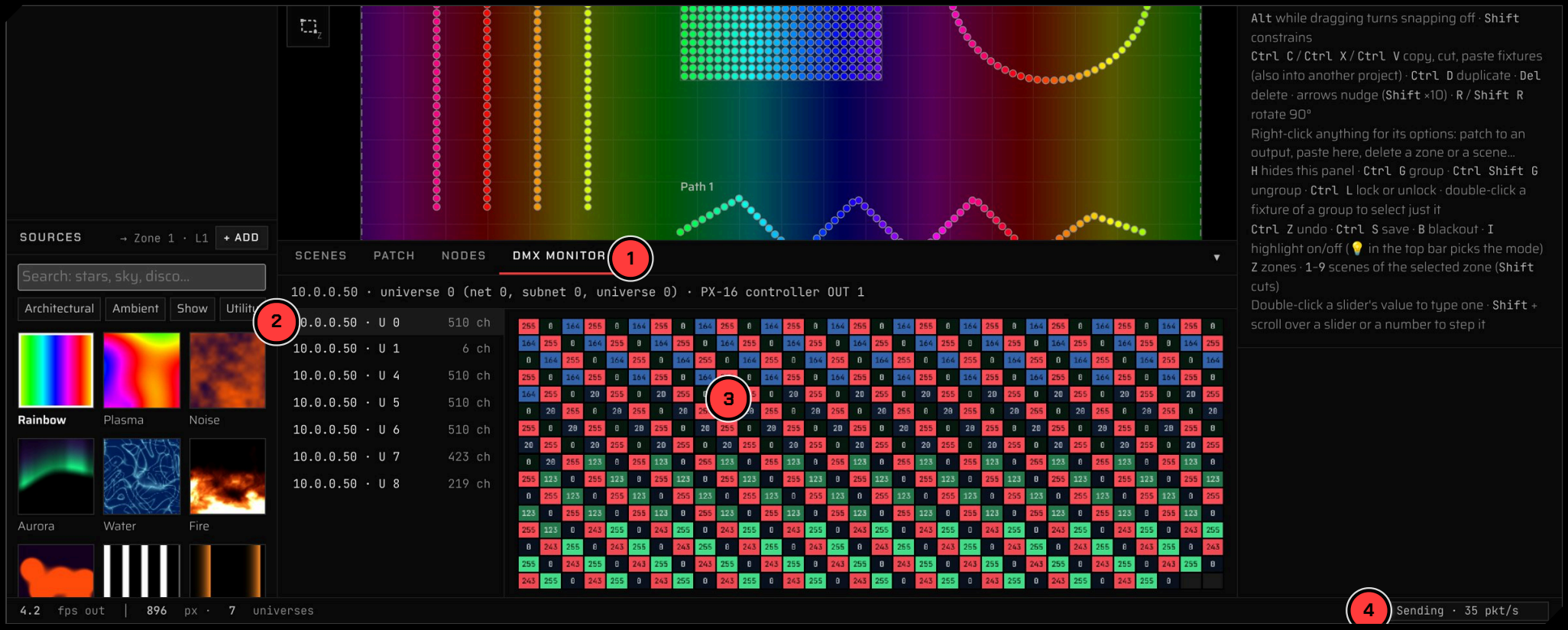
TIP Right-click a fixture → Patch to… The badge next to each fixture in the list shows its output.

10

DMX MONITOR

CHECK WHAT GOES OUT

1. **DMX monitor** tab: every universe that goes out, controller by controller.
2. **Pick a universe**: its 512 channels, live.
3. **Point at a channel** to see which fixture and pixel drive it.
4. Art-Net goes out from the start; the **status bar** shows how many packets a second.



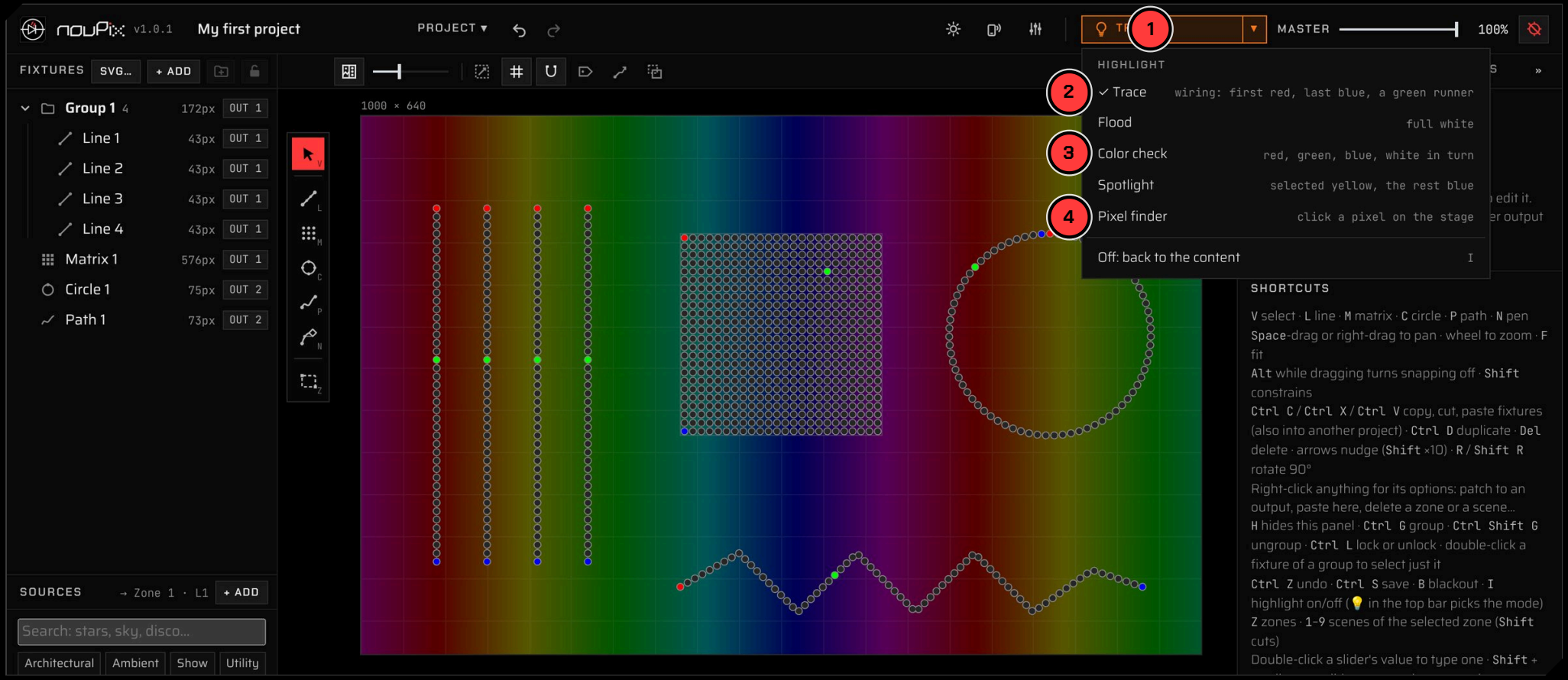
TIP Patched fixtures light the monitor even with no controller on the network: check the patch at your desk, before the site.

11

HIGHLIGHT ON SITE

CHECK EVERY FIXTURE

1. **Highlight** in the top bar, or `I`: the content stops and test patterns light the fixtures.
2. **Trace**: first pixel red, last blue, a green runner along the wiring.
3. **Flood, Color check, Spotlight** (the selected ones yellow, the rest blue).
4. **Pixel finder**: click a pixel on the stage to light only that one.



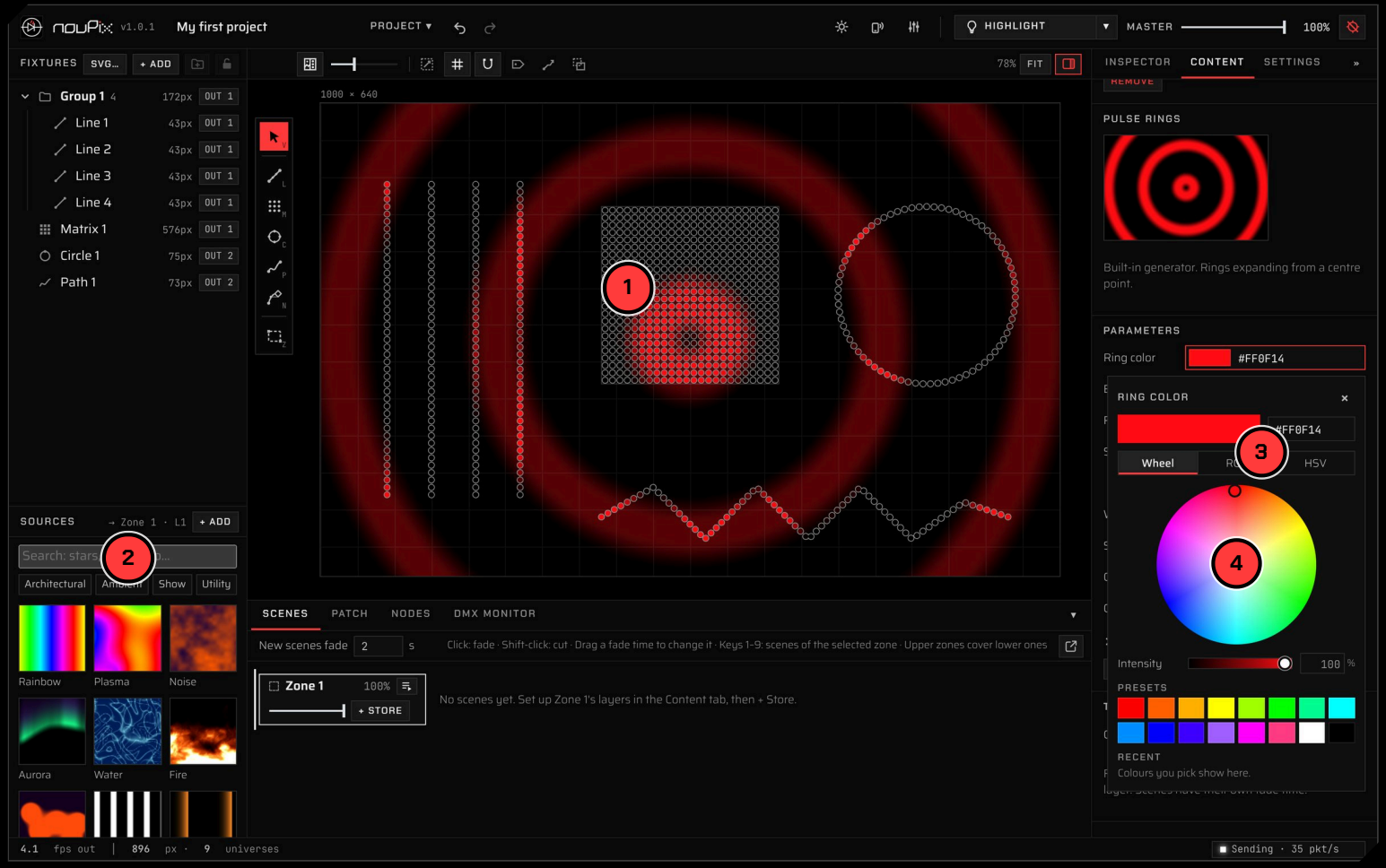
TIP Trace shows at once where a strip starts and whether it runs the right way.

12

PLAY CONTENT

CLICK A SOURCE, IT PLAYS

1. **Click a source**: it plays on the selected zone, and the stage shows it on the pixels.
2. **Search** or filter: Architectural, Ambient, Show, Utility.
3. **Its controls** in the Content tab. Double-click a value to type it.
4. **Colours**: wheel, RGB, HSV, presets and recent colours.



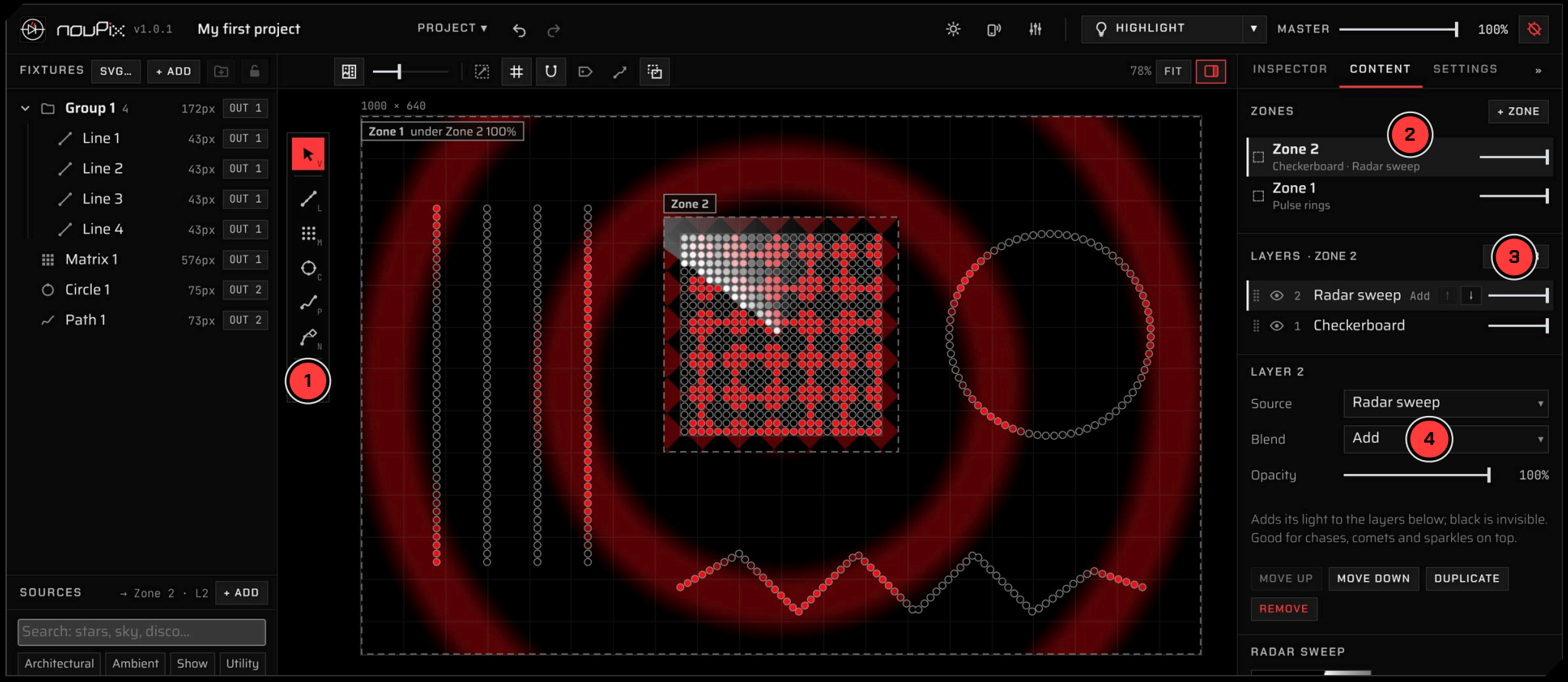
TIP `Shift` + scroll over a slider steps it; angles have buttons every 45°.

13

ZONES AND LAYERS

DIFFERENT CONTENT, DIFFERENT PARTS

1. **Z Zones tool**: drag a zone around part of the stage.
2. **Zones list**: the upper zone covers the lower. Drag to reorder.
3. **+ Layer** stacks another source in the zone.
4. **Blend** (Add, Screen, Multiply, Lighten…) and **opacity** per layer.



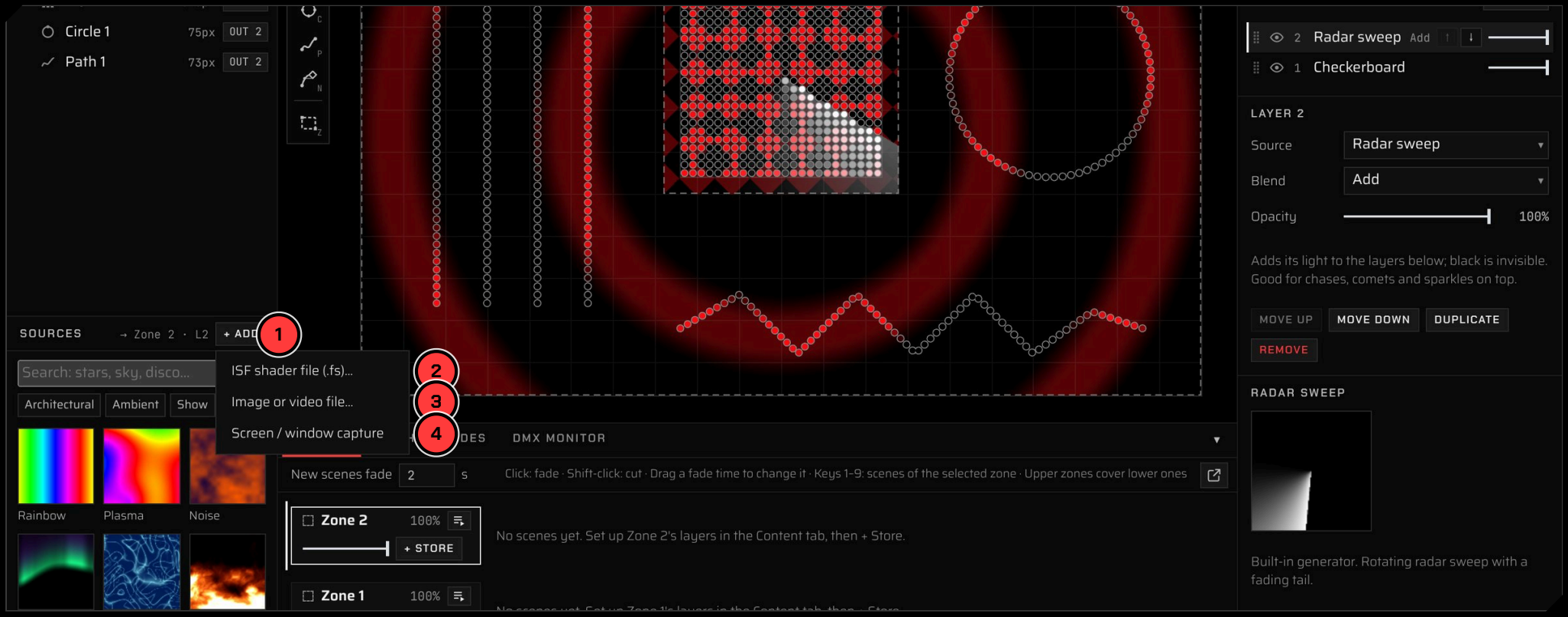
TIP Here Zone 2 plays a red checkerboard with a white radar sweep added on top, over the matrix only.

14

YOUR OWN CONTENT

SHADERS, VIDEO, CAPTURE

1. **+ Add** above the sources.
2. **ISF shader file (.fs)**: your own generators, with their controls.
3. **Image or video**: play, pause, trim, loop or once, with or without sound.
4. **Screen / window capture**: for example the output of other VJ software.



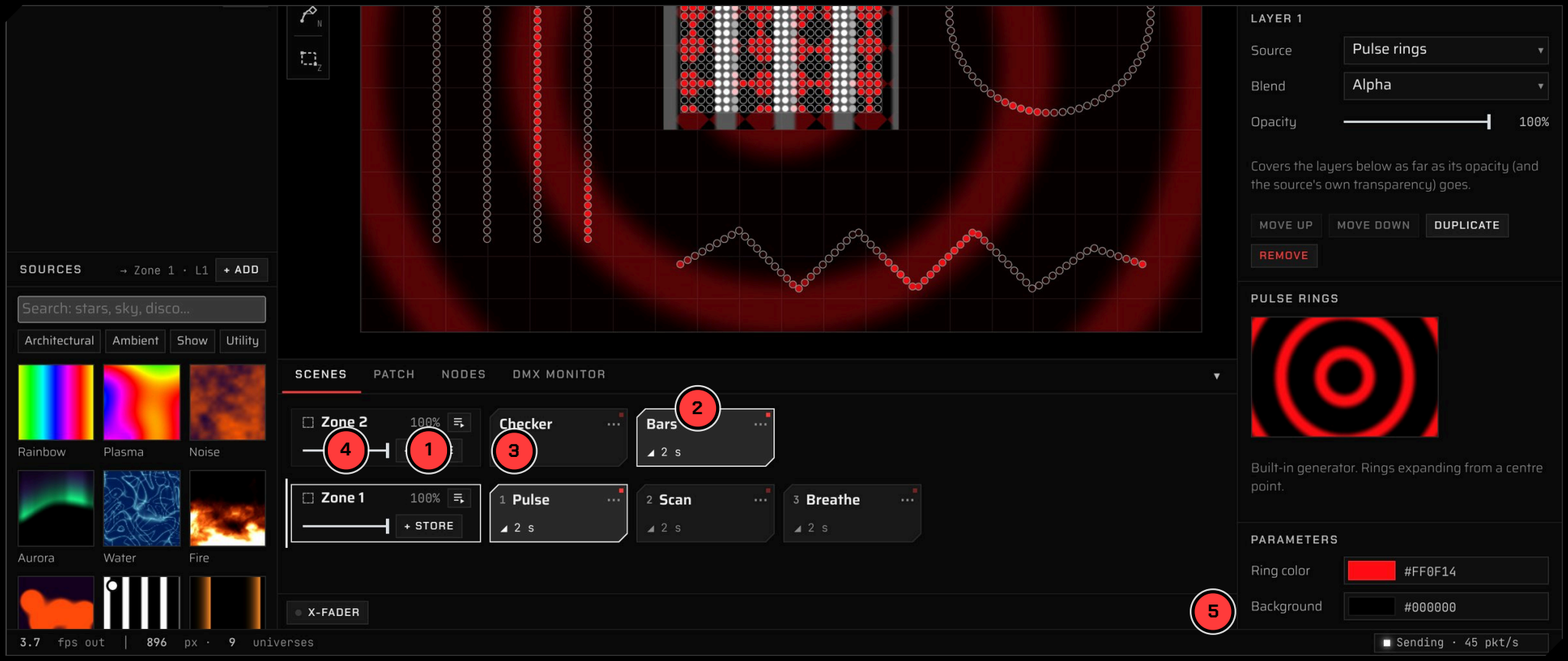
TIP Images and videos are not stored in the project file: bring them along, and re-link them from the Content tab on the show computer.

15

SCENES

STORE A LOOK, FADE BACK TO IT

1. **+ Store** keeps the zone's layers as a scene.
2. **Click a scene** to fade to it; `Shift`-click cuts.
3. **Drag the fade time** on its button to change it.
4. **Level** of each zone, live.
5. Open the **Scenes window** for a second screen.



TIP Keys `1`–`9` play the scenes of the selected zone.

16

X-FADER

ONE FADER BETWEEN TWO ZONES

1 **X-Fader**, bottom left of the Scenes panel, turns it on. **Make both full size** when asked: A and B then cover every light.

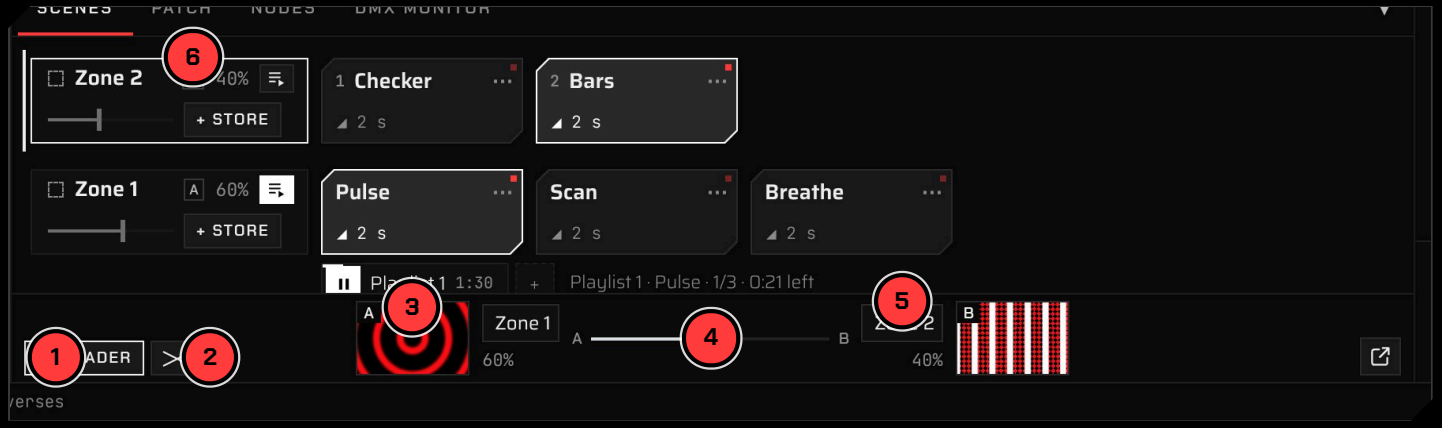
2 **Mode**: Mix, Dip through black, or Cut at the middle.

3 **Live previews** of A and B, even the one faded out.

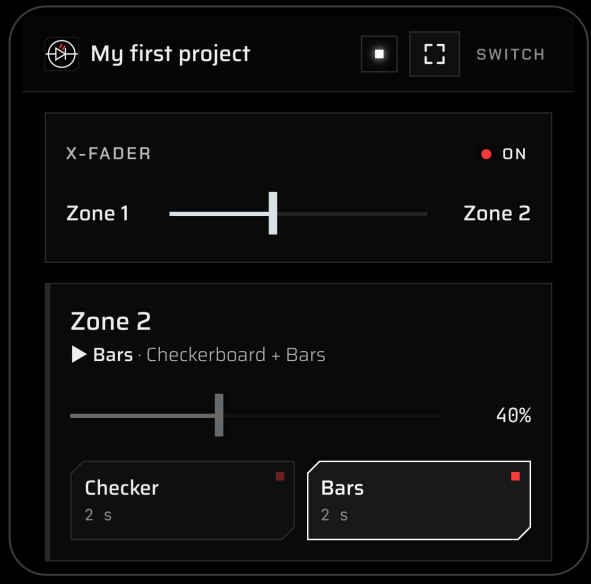
4 **The fader**: all A on the left, all B on the right.

5 **Choose the zones** for A and B from their lists.

6 **A** and **B** tags: while it is on, only the X-Fader sets their levels.







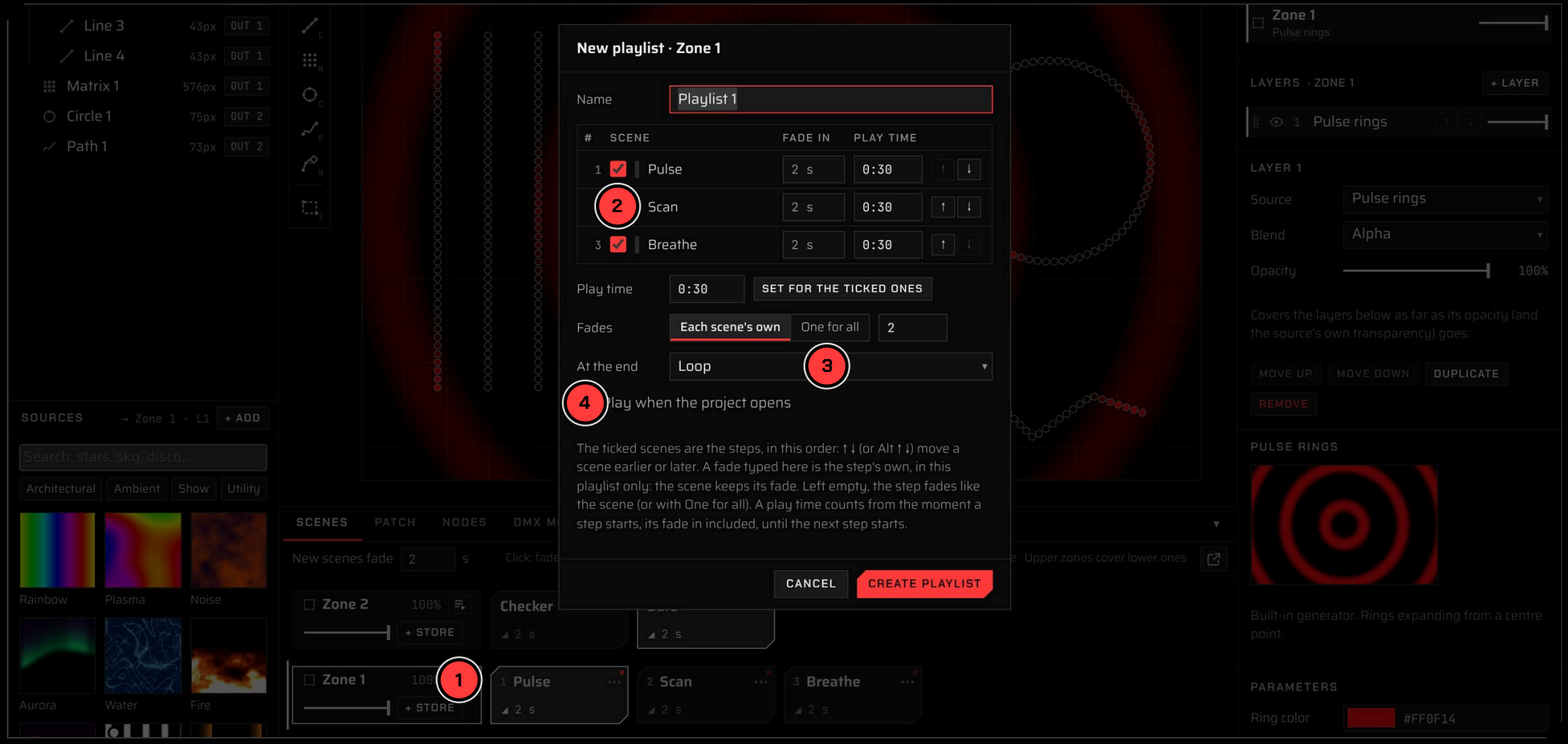
TIP Learn a MIDI fader onto it like any slider. Settings → Remote control → **X-Fader on the Remote page** puts it on the phones too.

17

PLAYLISTS

A SHOW THAT RUNS BY ITSELF

1. The **playlist button** next to + Store opens **New playlist**.
2. Pick the scenes, their play times and fades.
3. Play, pause, next. In a loop, once, or handing over to another playlist.
4. Tick **Play when the project opens**: the show starts by itself.



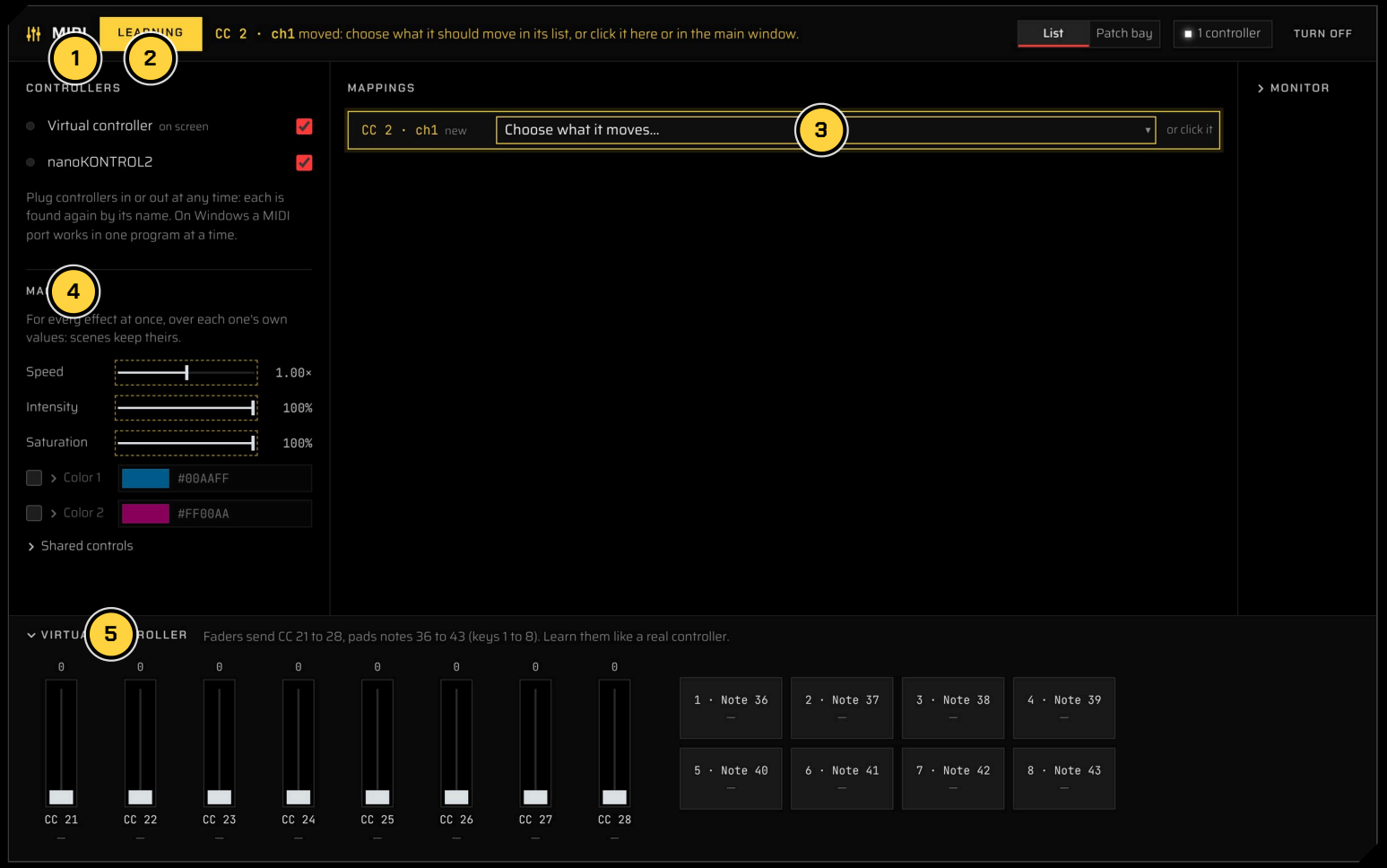
TIP A zone can have several playlists; the phone remote plays them too.

18

MIDI

LEARN YOUR CONTROLLER

1. **MIDI** in the top bar opens the MIDI window, with your controllers.
2. **Learn**, then move a fader, knob or pad.
3. **Choose what it moves** from the list, or click it in nouPix. Faders pick up where the slider is.
4. **Macros**: Speed, Intensity, Saturation and two colours, for every effect at once.
5. **Virtual controller**: faders and pads on screen, to try it without hardware.



TIP Master, blackout, the X-Fader, group dimmers, zone levels, scenes, playlists, macros and any control of an effect can be learnt.

19

PATCH BAY

EVERY MAPPING AS A WIRE

1

Patch bay: the same mappings, drawn as yellow wires.

2

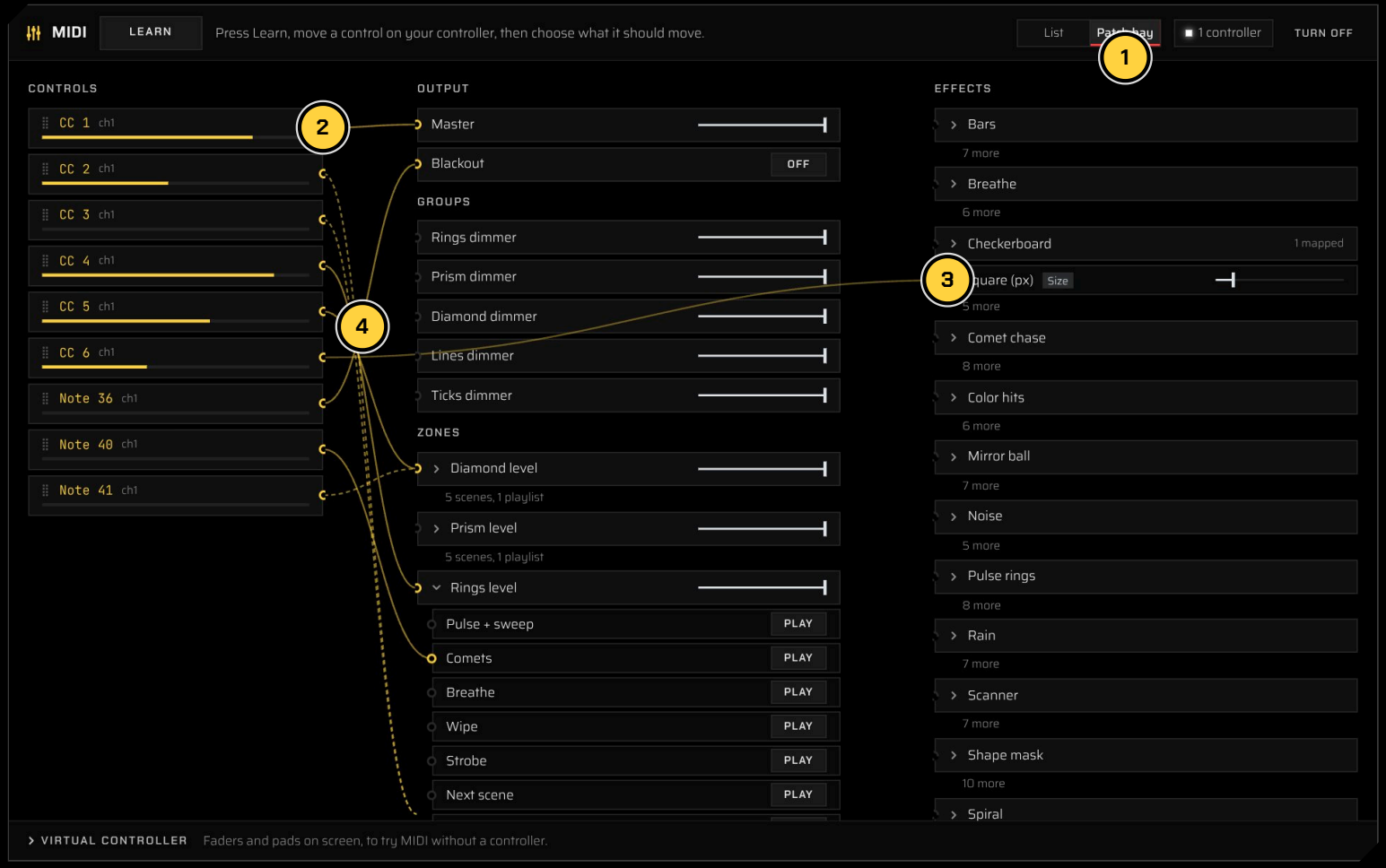
Controls on the left, **what they drive** in the middle: master, groups, zones, scenes, macros.

3

Effects on the right: wire any control of any effect.

4

Drag from a dot to another to wire them. Click a wire for its settings; Delete removes it.



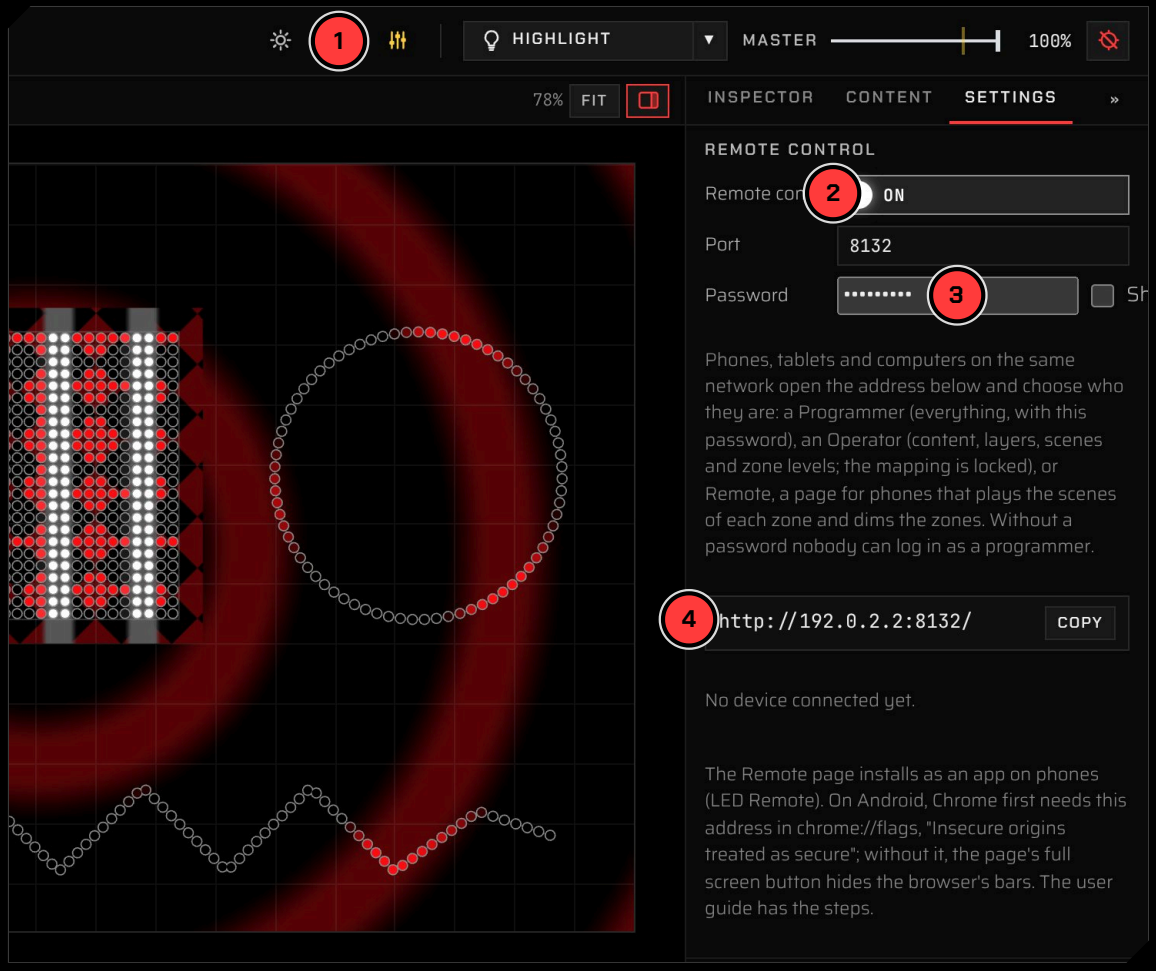
TIP Point at a macro to see, in aluminium wires, the effect controls that follow it.

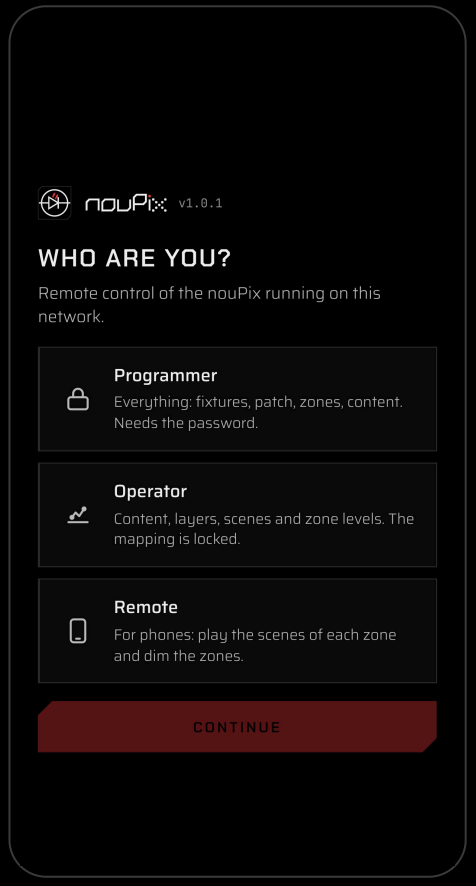
20

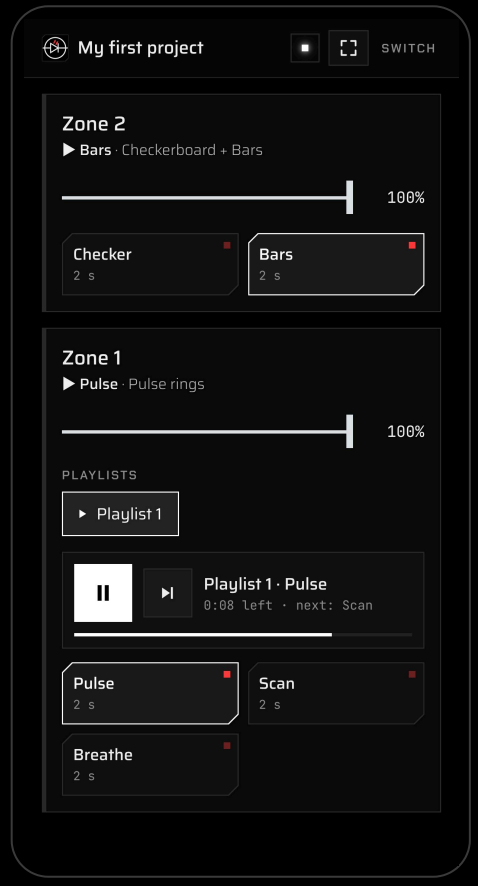
REMOTE CONTROL

HAND A PHONE TO THE END USER

1. **Remote** in the top bar → **Switch remote control on**, or Settings → Remote control.
2. **On**: a second server opens on your network.
3. Set a **password** for the Programmer role.
4. Open the **address** on a phone on the same network. **Remote** plays scenes and playlists, dims zones and, if you tick it, moves the X-Fader; **Operator** changes content too; **Programmer** everything.





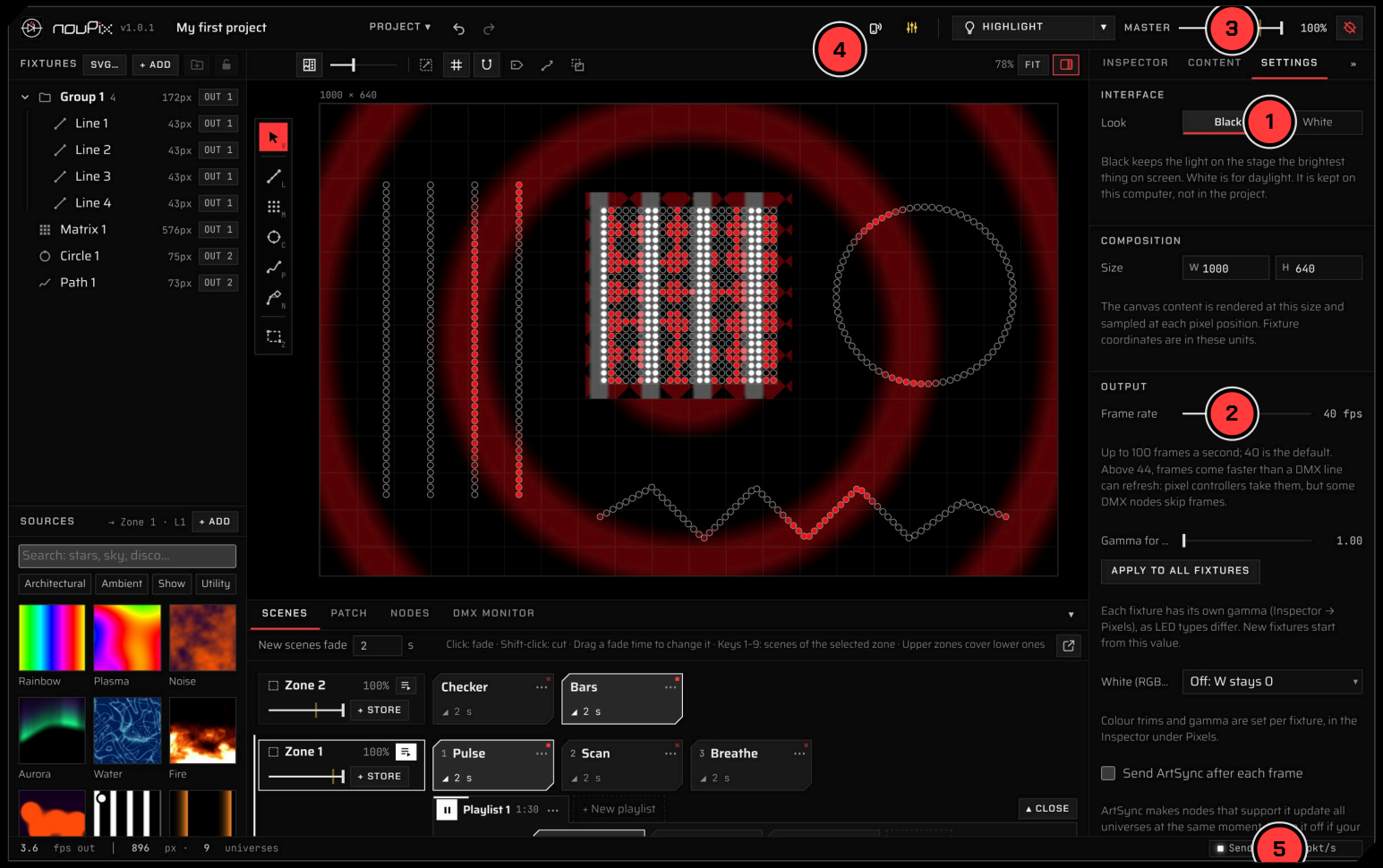


21

OUTPUT AND SETTINGS

MASTER, LOOK AND ART-NET

1. **Look**: black for the show, white for daylight.
2. **Frame rate** of the output, gamma, RGBW white.
3. **Master** and **blackout** (B) in the top bar.
4. The **light / dark** button switches the look at once.
5. **Art-Net**: on from the start; the status bar shows what goes out.



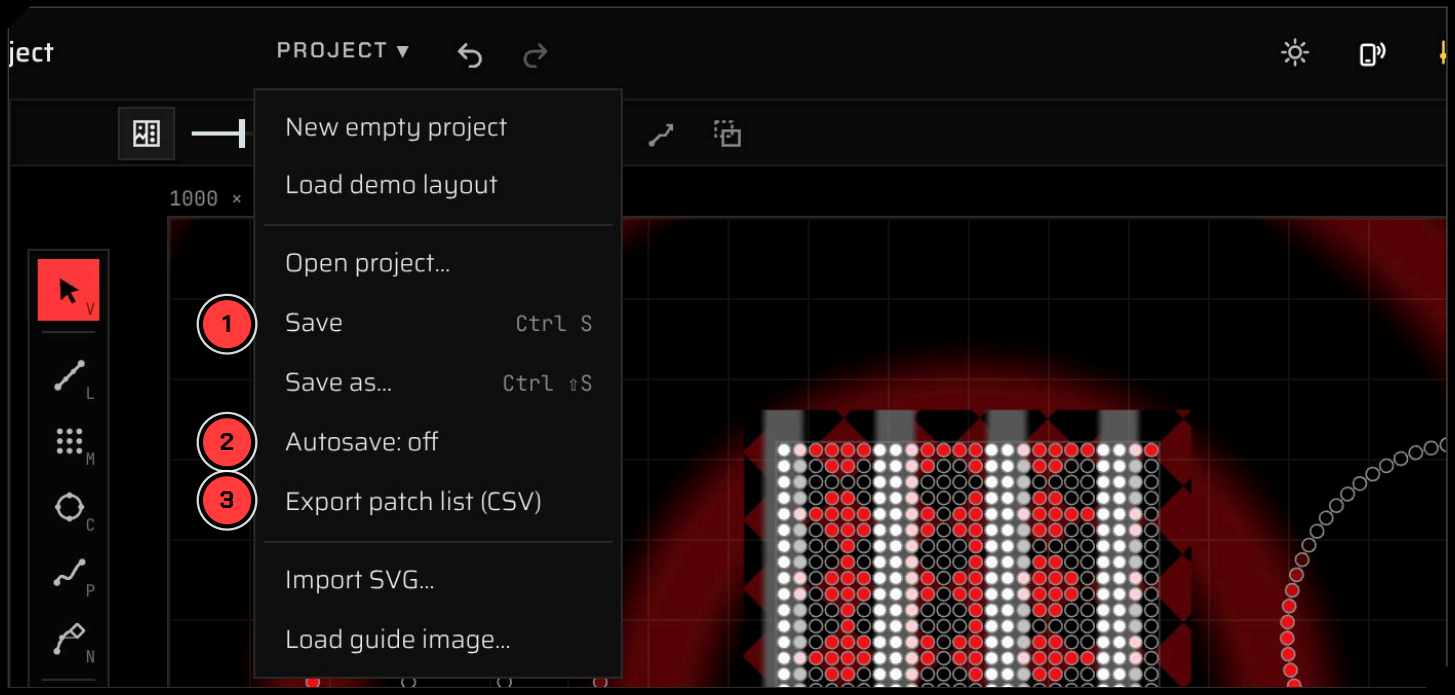
TIP Settings → Art-Net connection switches the output off, for example while a lighting console drives the lights.

22

SAVE AND HAND OVER

KEEP IT, SHARE IT

1. **Save** (`Ctrl` `S`) writes over the .noupix file; **Save as…** makes a new one.
2. **Autosave** writes every change to the file.
3. **Export patch list (CSV)**: universes and addresses for the install crew.



TIP The project file holds the fixtures, patch, zones, layers, scenes, playlists and MIDI. Images and videos stay separate files.

SHORTCUTS

KEYS TO REMEMBER

V	Select	L	Line	M	Matrix
C	Circle	P	Path	N	Pen
Z	Zones	Ctrl C	Copy	Ctrl X	Cut
Ctrl V	Paste	Ctrl D	Duplicate	Del	Delete
Ctrl G	Group	Ctrl Shift G	Ungroup	R	Turn 90°
Ctrl L	Lock	Ctrl Z	Undo	Ctrl S	Save
1-9	Scenes of the zone	I	Highlight	B	Blackout
F	Fit the view	H	Hide the side panel	Space	Drag to pan

On a Mac, Ctrl is ⌘.

DOWNLOAD

noupix.nouled.com

SOURCE AND MANUAL

github.com/nouled-studio/nouPix

BUGS AND IDEAS

github.com/nouled-studio/nouPix/issues

nouPix is free software under the GNU AGPL-3.0, made by NOULED.STUDIO and provided as it is, with no warranty. Art-Net™ designed by and copyright Artistic Licence Holdings Ltd.